

43/50 TOTAL	86.0% PERCENTAGE	A GRADE	8/10 SECTION A	20/24 SECTION B	15/16 SECTION C
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The short version. Lead with the good news: **Unit 6 (angle measurement and sector of a circle), the one part of this paper not yet taught in class, came back a perfect 11/11.** Seemab worked out radian–degree conversions and sector formulas entirely on her own, ahead of the classroom. Seemab scored 43 out of 50 overall (86.0%, Grade A). Question 8, the 8-mark grouped-data question, was a fully self-checking 8/8, with mean, median and mode all landing on exactly 25.5 as they must. Question 9, the construction, scored 7/8: the compass technique (angle bisectors and the in-centre) is sound, but the ruler-drawn base came out about a centimetre short, discussed in full below. Section A held at 8/10, with two confusable-pair mix-ups. The real story of this paper is Question 6, the marbles probability question, which does not appear anywhere in her notebook: not one part attempted. It is worth 4 marks on its own, more than every other single loss on the paper combined, and it happens to be the very first probability question (Unit 11, section 11.3) she has ever been set in any maths paper, home or school. Tellingly, she answered both probability MCQs (9 and 10) correctly, so the topic itself is not the barrier; it looks like the written, free-response format that tripped her.

Unit by unit: Unit 6 perfect at 11/11 (100%). Unit 10 (practical geometry), taught at school but never examined by us before, held at 13/15 (86.7%), all of it the Q.9 ruler slip plus one MCQ. Unit 11 (basic statistics), also taught but with probability appearing for the first time, is where the paper's marks were lost, at 19/24 (79.2%).

UNIT-BY-UNIT BREAKDOWN

UNIT	STATUS	MCQS	WRITTEN	TOTAL	%
Unit 6: Angle measurement & sector of a circle (6.1, 6.2)	Not yet taught in class	3/3 (Q1–3)	8/8 (Q.2, Q.3)	11/11	100%
Unit 10: Practical Geometry	Taught, never examined before	2/3 (Q4–6)	11/12 (Q.7, Q.9)	13/15	86.7%
Unit 11: Basic Statistics	Taught; probability (11.3) examined for the first time	3/4 (Q7–10)	16/20 (Q.4, Q.5, Q.6, Q.8)	19/24	79.2%

Unit 11's dip is entirely the untested-probability gap (Question 6, 4 marks) plus one mode/class-mark mix-up (1 MCQ mark). Every other Unit 11 item, including the full 8-mark grouped-data question, was correct. Unit 10's dip is one MCQ mix-up plus the single Q.9 ruler-accuracy mark, not a method problem.

QUESTION 6: NOT ATTEMPTED (0/4, the single largest loss on this paper)

The marbles probability question does not appear anywhere in her 8-page notebook. Her working runs Q.2, Q.3, Q.4, Q.5 (finishing with mode = 6), then goes straight to "Q:7" a couple of lines later on the very same physical page 4, with no gap. Page 8 (the MCQ page) has blank ruled lines below the answer list with nothing written on them. This continuous Q.5-to-Q.7 run is the strongest evidence for a genuine skip.

One honest caveat. The bottom line of page 7 is clipped by the CamScanner crop: her Q.9(iii) sentence runs off the edge mid-phrase ("lies inside the [triangle]"), and page 8 then opens cold with the MCQ list. A page, or a page foot, between the two cannot be completely ruled out on the scan alone. The evidence still points to a genuine skip, because Q.5 and Q.7 sit with no gap on page 4, so it is recorded as **not attempted, 0/4**. A re-scan of page 7 and anything after it would settle this for good, and if Q.6 turns up on it, the ceiling for this paper becomes 47/50.

Because section 11.3 (probability) has never once appeared on any of her maths papers before exam-19, and she answered both probability MCQs (Q.9 and Q.10) correctly on this very paper, the gap looks like an unfamiliarity with the written, multi-part format rather than the underlying idea. Correct working, for revision:

Total outcomes = 5 red + 3 green + 2 blue = 10
(i) P(green) = 3/10
(ii) P(blue) = 2/10 = 1/5
(iii) P(not red) = 1 – P(red) = 1 – 5/10 = 5/10 = 1/2
(iv) Expected number of 4s in 300 rolls = n × p = 300 × 1/6 = 50 times

If she attempts this question on a separate sheet that was not part of this scan, or a re-scan of page 7 onward turns up a missed page, the mark can be revised on sight of it.

SECTION A: Multiple Choice (8 / 10)

Q	QUESTION (SUMMARY)	CORRECT	SEEMAB'S	RESULT	WHAT HAPPENED
1	Seconds in one minute of an angle	C	C	1	Correct.
2	1 radian ≈	A	A	1	Correct (57°17'45").
3	Area of a sector, A = ?	D	D	1	Correct (A = ½r²θ).
4	Where the three angle bisectors meet	B	B	1	Correct (in-centre).
5	Right bisectors of a right-angled triangle	C	A	0	Wrong. She picked (a) "vertex of the right angle", which is where the <i>altitudes</i> of a right triangle meet. The key itself flags this as "the twin students most often pick". Correct answer is (c) "mid point of the hypotenuse".

6	Altitudes of an obtuse-angled triangle	D	D	1	Correct (outside the triangle).
7	Class mark of the class 5 to 15	B	B	1	Correct (10).
8	The most frequent value is called its	C	D	0	Wrong, but almost certainly a reading slip, not a knowledge gap. She picked (d) "class mark" here, yet answered "mode = 6" correctly of her own accord in Q.5(iii), later on this very script. She plainly knows what a mode is; something about reading the four options in Q.8 made her pick the wrong label. Correct answer is (c) "mode": "the most frequent (repeated) value of a data".
9	Probability of an impossible event	A	A	1	Correct (0).
10	Expected heads in 400 tosses	B	B	1	Correct (200).

Both MCQ losses are confusable-pair mix-ups, not blind guesses: Q5 confuses where *altitudes* meet with where *right bisectors* meet (both are correct facts about a right triangle, just answering different sub-questions), and Q8 confuses *mode* with *class mark* (both are legitimate statistics terms from the same unit, just different definitions, and she used "mode" correctly two questions later). Every other MCQ, across all three units, was answered correctly and confirmed at a high-resolution crop before being recorded. Worth noting: both probability MCQs (Q9 and Q10) were answered correctly, the same topic she left blank in Q.6.

SECTION B: Short Questions (20 / 24)

Q.2: Angle conversions (Unit 6)

4/4

(i) Convert 75° into radians: 2/2

75 × π/180 = 5π/12

(ii) Convert 5π/6 radians into degrees: 2/2

5π/6 × 180/π = 150°

Both parts correct, fully worked, correct units (degree sign present on 150°). Habit note, no mark attached: "5π/12" in part (i) carries no unit of its own, and the key's own wording awards that mark for "5π/12 in lowest terms" with no unit condition, so nothing is owed there either way.

Q.3: Arc length and area of a sector, r = 6 cm, θ = π/3 (Unit 6)

4/4

(i) Length of the arc: 2/2

L = rθ = 6 × π/3 = 6.28 cm

(ii) Area of the sector: 2/2

A = ½r²θ = ½ × 36 × π/3 = 18.85 cm²

Full marks, no deduction. The question specifies "taking π = 3.14" (key answer 18.84 cm²), but Seemab used the more precise π = 3.1416, giving 18.85 cm² (true value 18.8496). Her formula, substitution and method are all correct, and 18.85 is the more accurate answer, not an error. **Habit note only:** when a paper states a value of π to use, use that exact value so the final figure matches the key line for line: no mark was at stake here, but it is worth building the habit for board exams where a marking scheme may be stricter.

Q.4: Class 15 to 24: limits, boundaries, class mark, interval (Unit 11)

4/4

PART	HER ANSWER	KEY	MARK
(i) Lower / upper class limit	15 / 24	15 / 24	1/1
(ii) Class boundaries	14.5 – 24.5	14.5 – 24.5	1/1
(iii) Class mark	19.5	19.5	1/1
(iv) Class interval h	10 (from boundaries, 24.5–14.5)	10	1/1

She drew and abandoned a rough table for this question on the previous page, crossed it out, and restarted it properly here. Only the completed version is marked; no penalty for the abandoned attempt.

Q.5: Mean, median, mode of 4, 6, 9, 6, 10, 7, 6, 12 (Unit 11)

4/4

(i) Mean: 2/2

$$(4+6+9+6+10+7+6+12) / 8 = 60/8 = 7.5$$

(ii) Median: 1/1

Ascending order: 4, 6, 6, 6, 7, 9, 10, 12 → (4th + 5th)/2 = (6+7)/2 = 6.5

(iii) Mode: 1/1

mode = 6

Data correctly arranged in ascending order before taking the median: exactly the step the key warns is needed for full marks. All three values correct.

Q.6: Probability, bag of marbles (Unit 11)

0/4

Not attempted. See the highlighted section above for the full detail and the correct working. Recorded as a skip, not a wrong answer.

Q.7: Lines inside a triangle (Unit 10)

4/4

(i) Define the altitude: 1/1

"A perpendicular line segment drawn from a vertex of a triangle to the opposite side."

(ii) Define the median: 1/1

"A line segment joining a vertex of a triangle to the midpoint of the opposite side."

(iii) Point where the altitudes meet: 1/1

"The three altitudes meet at orthocentre"

(iv) Ratio the centroid divides each median: 1/1

"The centroid divides each median in ratio := 2:1"

SECTION C: Long Questions (15 / 16)

Q.8: Grouped data: mean, median, mode of 30 students' marks (Unit 11)

8/8

(i) Completed table (class boundaries, class marks x, fx, c.f.): 2/2. Every row matches the key: class marks 5.5, 15.5, 25.5, 35.5, 45.5; Σfx = 765; cumulative frequencies 4, 10, 20, 26, 30.

(ii) Arithmetic mean: 2/2

$$\bar{x} = \Sigma fx / \Sigma f = 765 / 30 = 25.5 \text{ marks}$$

(iii) Median: 2/2

$$n/2 = 15 \rightarrow \text{median group } 21-30 \text{ (l=20.5, h=10, f=10)}$$
$$\tilde{x} = 20.5 + (10/10)(15-10) = 20.5 + 5 = 25.5 \text{ marks}$$

(iv) Mode: 1/1

Modal class 21–30 (highest f = 10). Mode = 20.5 + [(10–6)/(20–6–6)] × 10 = 20.5 + 5 = 25.5 marks

(v) Concluding sentence: 1/1

"This means, mean, median and mode are all equal to 25.5, showing that the data is approximately symmetric and centered around 25.5 marks."

Self-check passed. The question is designed so mean, median and mode all land on exactly 25.5; all three of her answers agree, confirming every step along the way is correct. A genuinely excellent, fully worked answer. Habit note, no mark attached: her Q.8(v) sentence itself says "around 25.5 marks", which supplies the unit for the question, so nothing is owed there either.

(i) Triangle and steps of construction: 2/3

"Draw a line segment $AB = 6\text{cm}$. With B as a centre and radius 5cm , draw an arc. With A as a centre and radius 4.5cm , draw another arc cutting the first arc at C . Join AC and BC , then ABC is constructed."

One accuracy mark lost: the drawn base is about a centimetre short. Two independent measurements against the scan, one by the marking session and one by a blind second marker, both at 300 DPI on a 1 cm grid, agree: her three sides come out in the ratio $AB:BC:CA = 0.99:1:0.89$, where the paper needs 1.2:1:0.9. CA against CB checks out (0.89 vs 0.9), but AB is about 17% short: scaling off CB gives $AB = 4.95\text{ cm}$, scaling off CA gives $AB = 5.00\text{ cm}$, and reading the grid directly gives $AB = 5.09\text{ cm}$. The base was drawn at roughly 5 cm rather than 6 cm, a full centimetre outside the paper's 2 mm tolerance. Because AB and BC both came out close to 5 cm, she ended up drawing an isosceles triangle rather than the intended scalene one. Her written step 1 correctly says "Draw a line segment $AB = 6\text{cm}$ ", so this reads as a ruler slip when actually drawing the line, not a method error, and she keeps the steps mark and one of the two accuracy marks (the compass arcs at C are correctly struck and still on the page): 2/3. This is the single mark on the paper most open to challenge; the measurement comes from a dewarped CamScanner photo, and a ruler held to the original notebook page would settle it for good. She is entitled to ask for that recheck.

(ii) Bisectors of $\angle A$, $\angle B$, $\angle C$, meeting at O : 3/3

All three angle bisectors drawn with visible compass arcs at every vertex, converging to a single interior point. Her O sits about 1 mm from the true incentre of the triangle she actually drew (the 5, 5, 4.5 cm one, not the intended 6, 5, 4.5 cm one), so the bisector technique itself is sound; the earlier ruler slip does not carry over into this part.

(iii) Conclusion: 2/2

"Yes the three angle bisectors are concurrent. O is incentre of triangle ABC and lies inside the [triangle]."

Correct on every count: concurrency stated, point named as the in-centre, and O is drawn genuinely **inside** her triangle, exactly where it must fall for an acute triangle. A sound construction, let down only by the one measured side.

WHERE THE 7 LOST MARKS WENT

ITEM	MARKS	WHAT HAPPENED
MCQ 5 (Unit 10)	1	Wrote "vertex of the right angle" instead of "mid point of the hypotenuse".
MCQ 8 (Unit 11)	1	Wrote "class mark" instead of "mode" (she used "mode" correctly two questions earlier).
Q.9(i) (Unit 10)	1	Base AB drawn at about 5 cm, not the required 6 cm; compass technique and steps otherwise correct.
Q.6 (Unit 11)	4	Not attempted: her first-ever probability question.

Every other mark on the paper, all 43 of them, was earned with correct, fully-shown working. Nothing beyond the one measured side was lost to arithmetic slips, missing units, or an unfinished sentence anywhere in Sections B or C.

STRENGTHS

- Unit 6 (angle measurement and sector of a circle), the portion not yet covered in class, came back a flawless 11/11. Radian–degree conversions and the arc/sector formulas were applied correctly and independently, ahead of the classroom.
- Question 8, the 8-mark grouped-data question, self-checked cleanly (mean = median = mode = 25.5), a fully correct 8/8.
- Question 9's compass work: the angle bisectors and the in-centre are constructed correctly, and O lands inside the triangle exactly where it must, even though the ruler-drawn base itself was short.
- Every written definition (altitude, median, orthocentre, centroid ratio) was stated in her own accurate words.
- Class boundaries, class marks and cumulative frequency were all built correctly across two separate questions (Q.4 and Q.8), with the harder grouped-data version done without error.

AREAS TO REVISE

- Probability (Unit 11, section 11.3):** this is genuinely new ground, first tested on this paper and left blank entirely, even though both probability MCQs on the same paper were answered correctly. Needs a dedicated revision pass on the written, multi-part format: $P(E) = \text{favourable/total}$, complementary events $(1 - P)$, and expected frequency $= n \times p$.
- Right bisectors vs altitudes of a right-angled triangle:** two different points of concurrency, two different facts, and she has now mixed them once. Worth a quick side-by-side drill of all four points of concurrency (in-centre, orthocentre, circumcentre, centroid) and which construction produces which.
- Mode vs class mark:** a reading slip rather than a knowledge gap. She answered "mode = 6" correctly herself, unprompted, in Q.5(iii) on this very script, so the definitions are known; the four printed options in Q.8 just got misread.
- Measure the base before moving on:** Q.9's steps and compass work were both right, but the ruler-drawn AB came out close to a centimetre short, enough to turn the intended scalene triangle into an isosceles one. A quick check of each side against the ruler straight after drawing it would catch this before the bisectors are even started.
- When a paper states a value for π , use that exact value** so the final answer matches the marking key line for line, even when a more precise value would also be correct.

COMPARISON WITH PREVIOUS MATHS ATTEMPT

EXAM	DATE	SCORE	%	GRADE
exam-18-ch1-2-4-5-7 (Version B)	15 Sep 2026	32.5/38	85.5%	A
exam-19-ch6-10-11 (this attempt)	23 Sep 2026	43/50	86.0%	A

A small improvement over exam-18 (+0.5 percentage points), on a paper covering entirely new territory: two units (10 and 11) never examined by us before, and one unit (6) not yet taught in class at all. Given that mix, holding above 86% and posting a perfect Unit 6 score is a solid result, and probability is now the clear, specific target for the next paper.

Marking method. All 8 pages of the CamScanner scan were read as page images at 200 DPI; the MCQ list (page 8) and the Q.3(ii) area unit were re-checked at 200–

250% crop zoom before being recorded. Every question, sub-part and MCQ option was checked directly against the exam paper and worked from the answer key's method, not just compared to the printed answer. Question 6's absence was investigated specifically: the notebook runs fully sequentially from Q.2 through Q.9 with no page-break gap on page 4 (where Q.5 ends and Q.7 begins), though the clipped last line of page 7 means a page between Q.9 and the MCQs cannot be fully ruled out from the scan alone; the evidence still points to a genuine skip. Question 9's construction was independently re-measured at 300 DPI against a 1 cm grid by a blind second marker, who agreed with the finding that the base AB was drawn short; both measurements are reported above. This report reconciles both passes.

END OF MARKING REPORT